Block Three Synthesis. Biliary Atresia, From the First Bilirubin to the Adult Liver.

Imagine five patients who could all arrive at the same liver clinic in the same week. The first is a breastfed boy of eighty-two days whose parents mentioned pale stools only at his two-month vaccinations. The second is a four-month-old girl who had a good operation at forty-one days and now has a fever and a rising bilirubin. The third is a boy of four and a half months whose operation never drained, whose bilirubin is still above seven milligrams per decilitre, and who is falling off his growth curve. The fourth is a seven-year-old girl with a normal bilirubin, a big spleen, a low platelet count and an oxygen saturation that is slightly lower when she stands than when she lies down. The fifth is a woman of twenty-two who still has her own liver and wants to become pregnant.

All five have the same disease. Biliary atresia is a progressive, inflammatory and scarring destruction of the bile ducts outside the liver that begins before or around birth. Without surgery it is fatal within the first years of life. With the operation designed by Morio Kasai, called the Kasai portoenterostomy, about half of infants clear their jaundice, but most still need a liver transplant at some point, and the disease remains the single commonest reason for liver transplantation in children. The last thirteen lessons followed this disease from the newborn period to adulthood. This chapter brings them together and asks one practical question. What, in biliary atresia, can we still change?

The main idea of the block can be put in one sentence. Biliary atresia runs on two clocks. The first clock runs before the Kasai, and it runs in days. Every week that passes before the operation removes some of the tiny bile channels that the surgeon hopes to drain, and the age at which the infant reaches the operating theatre is set mostly by the health system: by whether anyone looks at the stool, measures the conjugated bilirubin, refers urgently, and sends the infant to a surgeon who does this operation often. The second clock starts after the Kasai and runs for the rest of the patient's life. A drained liver is still a scarred liver. It keeps developing fibrosis, so cholangitis, portal hypertension, abnormal blood vessel connections in the lungs, poor growth and eventually cancer must be looked for before they announce themselves.

The stakes are clear in the national registries. About half of infants clear their jaundice after the Kasai. In the Japanese registry, about four in ten patients still have their own liver at thirty years. Yet overall survival in good systems is above eighty-five to ninety percent, and that difference exists only because transplantation rescues the children whose Kasai fails. We will move through the material in the same order as the slides. First the mechanisms, then the guidelines and where they part, then the evidence on the two

clocks, then the five patients, and finally twenty pearls, the gaps in our knowledge, a short self-test, and what to do differently in clinic.

Part one. One disease, seven processes.

It helps to think of every problem in biliary atresia as one of seven processes, because each process has its own clue and its own first action.

The first process is the obliterative cholangiopathy itself. The current view is a multi-hit model. A susceptible bile duct, perhaps shaped by genetic variants or by abnormal development, suffers an injury before or soon after birth, possibly from a virus or a toxin, and the immune system then amplifies that injury and drives scarring until the duct is closed. The best clinical evidence that the disease starts this early is that the conjugated, or direct, bilirubin is already raised in the first two days of life in infants who will later be diagnosed with atresia. The clue in clinic is a pale stool, a high gamma-glutamyl transferase, which we will call GGT, and a firm liver. The first action is to measure the direct bilirubin in any infant who is jaundiced at two to four weeks, and to run the work-up in parallel rather than one test at a time.

The second process is the loss of the tiny bile channels at the porta hepatis, the hilum of the liver where the surgeon cuts across the scarred duct remnant. The Kasai does not reconnect a bile duct. It opens the fibrous plate at the porta so that bile from surviving microscopic ductules can drain into a loop of bowel. Each week of obstruction removes some of those ductules and adds fibrosis inside the liver. The clue is simply the age of the infant and the amount of fibrosis on a biopsy. The first action is a Kasai without delay in an experienced centre.

The third process is ascending infection, called cholangitis. The Roux loop of bowel that is joined to the porta gives gut bacteria a route into the drained ductules. The clue is subtle. In one series, fever was present in only about four in ten episodes that clinicians diagnosed as cholangitis, while the bilirubin or the GGT rose in about nine in ten. The first action is to take cultures and start broad-spectrum antibiotics, and to treat for ten to twenty-one days depending on how certain the diagnosis is.

The fourth process is progressive fibrosis, which continues even after excellent drainage and leads to portal hypertension. The clues are a falling platelet count, a growing spleen and an increasing liver stiffness. The first action is endoscopy when portal hypertension is suspected, and band ligation of high-risk varices once a child weighs about ten kilograms.

The fifth process is portosystemic shunting. When blood bypasses the liver, substances that the liver would normally clear reach the lungs. In some children the small lung

vessels dilate, causing the hepatopulmonary syndrome, which lowers the oxygen level. In others, usually adolescents, the lung arteries constrict and thicken, causing portopulmonary hypertension. The strongest clue for the hepatopulmonary syndrome is a splenic malformation, such as polysplenia, and the trap is that the oxygen saturation is often normal early on. The first actions are a contrast echocardiogram with an arterial blood gas, and an echocardiographic estimate of the right ventricular pressure.

The sixth process is malabsorption and catabolism. Without bile in the gut, the infant cannot absorb long-chain fats or the fat-soluble vitamins A, D, E and K, and the sick liver increases energy needs to roughly one hundred and thirty to one hundred and fifty percent of normal. The clue is growth. Six months after the Kasai, infants who later died or needed a transplant had a weight-for-age z score of about minus two point one, compared with about minus one point two in those who kept their native liver. The first action is to provide at least one hundred and thirty percent of the energy requirement, a formula rich in medium-chain triglycerides, which we will call MCT, tube feeding when needed, and vitamins dosed and measured one by one.

The seventh process is malignant change. A chronically injured, often cirrhotic liver can develop hepatocellular carcinoma, which in biliary atresia tends to appear in young children, and cholangiocarcinoma, which tends to appear in young adults. The first action is lifelong surveillance with ultrasound and alpha-fetoprotein every six months, and a cancer marker called CA 19-9 in adults.

Part two. The anchor documents.

Six documents anchor this block, and it is worth knowing what each one is and what it rests on.

The first is the 2017 guideline of the North American and European Societies for Paediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN and ESPGHAN, led by Rima Fawaz. It covers the evaluation of the cholestatic infant. It used the GRADE method and made ten recommendations, most of them based on low-quality evidence. The two lines to remember are that a direct bilirubin above one milligram per decilitre is abnormal at any total bilirubin, and that the intraoperative cholangiogram is the gold standard for the diagnosis.

The second is the 2025 clinical report of the American Academy of Pediatrics, led by Sanjiv Harpavat, which is aimed at finding atresia in primary care. It asks clinicians to measure the direct bilirubin in jaundiced infants at two to four weeks, to treat a value of one milligram per decilitre or more as urgent, and to aim for a Kasai before thirty days of

age.

The third pair are the transplant documents of the American Association for the Study of Liver Diseases, called AASLD. The 2014 guideline on the evaluation of children for liver transplantation, led by Robert Squires, advised referral when the bilirubin stays above six milligrams per decilitre beyond three months after the Kasai, and consideration from two milligrams per decilitre. The new 2026 joint guideline of the AASLD, the American Society of Transplantation and NASPGHAN, led by Vicky Ng, moves the emphasis earlier. It asks for referral at the first complication, such as growth failure, recurrent cholangitis, portal hypertension or severe itch, and it asks that every candidate be screened for the hepatopulmonary syndrome with pulse oximetry.

The fourth is the 2021 guideline of the Japanese Biliary Atresia Society, together with the Japanese Biliary Atresia Registry, which has collected nearly four thousand patients since 1989. The guideline is notable for its honesty. On steroids after the Kasai it makes no recommendation either for or against, and on transplant indications it declined to recommend anything for lack of evidence.

The fifth is the English centralisation programme, described by Mark Davenport and colleagues. Since 1999, every infant with suspected atresia in England and Wales has been sent to one of three centres that do both the Kasai and liver transplantation. Its twenty-year report, published in 2024, covers eight hundred and sixty-seven infants.

The sixth is the 2022 international Delphi consensus on care after the Kasai, from a group of experts in biliary atresia. It defined suspected and confirmed cholangitis, suggested treatment for ten to fourteen days when cholangitis is suspected and fourteen to twenty-one days when it is confirmed, and supported antibiotic prophylaxis for six to twelve months.

Part three. Agreed before the Kasai, silent after it.

When we compare these documents question by question, a clear pattern appears. Before the operation they mostly agree. After the operation they are mostly silent.

They agree on the bilirubin trigger. The North American documents treat a direct bilirubin above one milligram per decilitre as abnormal, and the English national guidance uses a conjugated bilirubin above twenty-five micromoles per litre, which is about one and a half milligrams per decilitre, at the prolonged jaundice check. That is aligned.

They differ on universal screening. The American Academy of Pediatrics calls universal screening a future goal. Taiwan, Japan and Switzerland use stool colour cards, and in

2026 the United Kingdom National Screening Committee still judged the evidence insufficient to recommend either a card or a blood test.

There is a material difference on a newer blood test called matrix metalloproteinase seven, or MMP-7. The 2017 guideline predates it and says nothing, while the 2025 Chinese guideline already recommends it, with a weak grade. There is another material difference on who should operate. The North American documents ask for an experienced centre but give no number, while England has concentrated the operation in three centres and the Nordic countries require more than three cases a year.

On steroids after the Kasai the documents differ, with Japan declining to recommend for or against and the North American documents not addressing the question. On cholangitis prophylaxis and on screening for varices in children there is simply a gap, because no society guideline speaks to them, and the only paediatric statement on varices, the paediatric report from the Baveno conference, found no consensus. On referral for transplant the documents differ: the 2014 AASLD guideline gave a bilirubin threshold, the 2026 guideline moved to the first complication, and Japan declined to recommend.

The material point is this. Every document wants an early and well-done Kasai, but no North American document requires either the screening or the centralisation that delivered earlier and better surgery elsewhere. And after the Kasai, the most commonly used drugs and the most common complication have no society guideline at all.

Part four. The block in one chart.

Each of the thirteen lessons ended with a table comparing the guidelines, and each answer in those tables was given one of four verdicts. When all eighty-three answers are counted, thirty, or thirty-six percent, were aligned. Twenty-two, or twenty-seven percent, differed. Seventeen, or twenty percent, were gaps where nobody says anything. And fourteen, or seventeen percent, were material differences that would change what happens to a child. In other words, only about one answer in three is agreed, and two in three are a disagreement or a silence.

The silences cluster after the operation: cholangitis, adjuvant drugs, varices in infants, the lung syndromes, surveillance of jaundice-free survivors, and biliary cancer in adults. The reason is structural. The AASLD and the European Association for the Study of the Liver, called EASL, write mainly for adults. The joint paediatric guideline of 2017 stops at diagnosis. And the best data in biliary atresia come from national registries, not from randomised trials.

Part five. The first clock.

The evidence for the first clock comes from national cohorts. In the French national registry of one thousand three hundred and forty operated infants, survival with the native liver at twenty-five years was thirty-eight percent when the Kasai was done in the first month of life, twenty-seven percent in the second month, twenty-two percent in the third month, and nineteen percent after three months. In the Canadian national cohort of three hundred and forty-nine infants, native liver survival at four years was forty-nine percent with a Kasai at thirty days or earlier, thirty-six percent between thirty-one and ninety days, and twenty-three percent after ninety days. The shape is the same in both systems. The youngest do best, and even the oldest keep their liver about one time in five.

A 2023 meta-analysis of nine studies put a modest size on the effect: a Kasai at thirty days or earlier increased native liver survival by about twenty-two percent in relative terms compared with a later operation. The size is modest, but the direction is consistent, and there is no age at which the curve flattens. There is no safe age to wait, and no age at which the operation becomes pointless.

Screening can move the clock. In Texas, Sanjiv Harpavat and colleagues measured the direct bilirubin in the routine newborn sample and again at the two-week visit in about one hundred and twenty-four thousand newborns. All seven infants with atresia were found, and the mean age at the Kasai fell from fifty-six to thirty-six days. In the real world, however, about forty-five percent of American infants still had their Kasai at sixty days or later between 2016 and 2019, and the median age at different children's hospitals ranged from thirty-eight to seventy-six days. That five-week spread is a systems problem, not biology.

Part six. The second clock.

The second clock is set partly by the centre and partly by time. In the United Kingdom and Ireland in 1993 to 1995, before centralisation, sixty-one percent of children operated in centres doing more than five Kasai procedures a year still had their native liver at five years, compared with fourteen percent in centres doing fewer. After England centralised the operation, the twenty-year report of eight hundred and sixty-seven infants showed a median age at Kasai of fifty-one days, jaundice clearance in about sixty-one percent, native liver survival of about fifty-one percent at five years and forty-seven percent at ten years, and overall survival of about ninety-one percent at five years.

The decades then bend the curve downward. In the Japanese registry of three thousand nine hundred and fifty-one patients, native liver survival was about fifty and a half percent at ten years, forty-four percent at twenty years and forty-one percent at thirty years. In France it was twenty-two percent at thirty years. Drainage is a reprieve, not a

cure.

The single best early predictor of the second clock is the bilirubin three months after the Kasai. In the North American Childhood Liver Disease Research Network, called ChiLDReN, one hundred and thirty-seven infants were followed. Half of them reached a total bilirubin below two milligrams per decilitre within three months. Their two-year survival without transplant was eighty-six percent, compared with only twenty percent in those whose bilirubin stayed at two or above.

Part seven. Six numbers that move decisions.

Six numbers from the block deserve a place in memory, because each one changes what happens next to a real child. The first is eighty-six percent versus twenty percent, the two-year transplant-free survival with a bilirubin below or above two milligrams per decilitre at three months. The second is sixty-one percent versus fourteen percent, the native liver survival at five years in high-volume and low-volume British centres.

The third is zero out of four, the number of adjuvant drugs that have worked after the Kasai. In the START trial, high-dose steroids gave bile drainage in about fifty-nine percent compared with forty-nine percent on placebo, a difference that was not statistically significant, with no difference in transplant-free survival and with more serious adverse events early on. Intravenous immunoglobulin in the PRIME study did no better than historical controls. Two drugs that block the intestinal reabsorption of bile acids also failed: maralixibat in the EMBARK study, and odevixibat in the BOLD trial, which randomised two hundred and fifty-four infants in nineteen countries and in 2026 did not improve survival with the native liver.

The fourth number is forty-one percent, the share of clinician-diagnosed cholangitis episodes in which the child actually had a fever. The fifth is sixty-two percent, the share of infants under twelve months with high-risk varices whose platelet count was one hundred and fifty thousand or more, in other words normal. The sixth is zero out of twenty-three, the number of infants with a bilirubin of two or more who were sufficient in all four fat-soluble vitamins six months after the Kasai. The common lesson is that every complication of atresia announces itself late, so we must look for it before it does.

Part eight. Five patients.

The first patient is the late referral. A breastfed boy of eighty-two days has had pale stools for a few weeks. His direct bilirubin is six point eight milligrams per decilitre, his GGT is six hundred and ten units per litre, and his albumin is thirty-six grams per litre. His international normalised ratio, the INR, was one point six and corrected to one point one

after vitamin K. The ultrasound shows a small irregular gallbladder, the echogenic triangular cord at the porta, a spleen of six and a half centimetres and no ascites. His parents ask whether it is too late. It is not. After three months, one in five French children still had their native liver at twenty-five years. His INR corrected with vitamin K and he has no ascites, so his liver synthesis is preserved. A primary transplant without a Kasai is reserved for the decompensated liver, not for the calendar. A working Kasai, even a partial one, buys growth and time to reach a safer transplant later. So he should have his cholangiogram and Kasai this week in an experienced centre, with early transplant assessment if the bilirubin does not fall. And someone should ask why the two-week visit missed him.

The second patient has a fever after the Kasai. A girl of four months had her Kasai at forty-one days, and her bilirubin had fallen to one point six. She is on co-trimoxazole prophylaxis. Today she has a temperature of thirty-eight point four degrees, she is irritable, her stools are paler, her bilirubin has risen to three point nine, her GGT has risen from four hundred and eighty to seven hundred and ninety, and her C-reactive protein is sixty-four milligrams per litre. Under the 2022 consensus, two clinical and two laboratory criteria, or a positive culture, mean confirmed cholangitis. Blood cultures are positive in only ten to forty-four percent of episodes, so a negative culture does not stop treatment. Prophylaxis does not exclude infection, and third-generation cephalosporins are failing more often because of resistant gut organisms. She should be admitted and given piperacillin with tazobactam, chosen according to local resistance, for fourteen to twenty-one days, escalating to meropenem if there is no response at seventy-two hours. An ultrasound should look for bile lakes, which, together with recurrent episodes, point towards transplant.

The third patient is the Kasai that did not drain. A boy of four and a half months had his Kasai at fifty-eight days, and his bilirubin never went below five. It is now seven point two. His weight-for-age z score is minus two, his arm circumference is falling, his vitamin D is twelve nanograms per millilitre, his vitamin E is low, and he has mild ascites. Both parents offer to donate. A bilirubin of two or more at three months predicts a two-year transplant-free survival of about twenty percent. The 2014 AASLD guideline would refer him because his bilirubin is above six, and the 2026 guideline would have referred him at his first complication, which was his growth. In the North American transplant registry called SPLIT, a height or weight more than two standard deviations below the mean predicted death and graft loss after transplant, so his nutrition is now part of his transplant preparation. In the American national transplant data for biliary atresia, five-year graft survival was about ninety-four percent with a living donor graft compared

with about eighty-one percent with a split graft. He needs transplant evaluation now, nasogastric feeding to at least one hundred and thirty percent of his energy needs with an MCT-rich formula, each vitamin corrected and then measured again, and living donor assessment of his parents started in parallel.

The fourth patient is the cleared child. A girl of seven had her Kasai at thirty-five days and cleared her jaundice by two months. She has polysplenia and an interrupted inferior vena cava. Her bilirubin is zero point eight, but her platelets are eighty-eight thousand and her spleen is four centimetres below the ribs. She tires on stairs, and her oxygen saturation is ninety-five percent upright and ninety-seven percent lying down. About half of children who survive with their native liver have portal hypertension, and a normal bilirubin does not exclude it. In children, non-invasive tests such as platelets and liver stiffness cannot replace endoscopy. Her splenic malformation is the strongest known risk factor for the hepatopulmonary syndrome, and a fall in saturation on standing is exactly the pattern that syndrome produces, because the dilated vessels sit mainly in the lung bases. Pulse oximetry finds only late disease. She needs a contrast echocardiogram, an arterial blood gas and an echocardiographic estimate of her pulmonary pressure, together with endoscopy and banding of any high-risk varices. If the hepatopulmonary syndrome is confirmed, it is an indication for transplant with good results, and it resolves after transplant.

The fifth patient is the adult. A woman of twenty-two had her Kasai at fifty days and moved to adult care at eighteen. She had two episodes of cholangitis at fifteen. Her bilirubin is one point four, her platelets are ninety-five thousand, her spleen is sixteen centimetres, and her MELD score, the model for end-stage liver disease, is eight. Her last endoscopy, three years ago, showed small varices. Her new adult hepatologist plans to follow her as if she had primary sclerosing cholangitis. Before conception, a MELD score below six predicts an excellent outcome and a score above ten predicts decompensation, so she sits in between. In a Japanese survey of sixteen women with twenty-three deliveries, the liver worsened after delivery in about thirty-eight percent and cholangitis occurred in a quarter. She needs an endoscopy within a year before conception, continuation of a beta-blocker if she is taking one, banding of high-risk varices, and delivery planned for obstetric rather than hepatic reasons. Above all, she does not have primary sclerosing cholangitis. She has a cirrhotic liver with a Roux loop, at risk of both hepatocellular carcinoma and cholangiocarcinoma, so she needs ultrasound and alpha-fetoprotein every six months, a CA 19-9, and a full cancer work-up for any new biliary stricture. Transplant should be revisited if she decompensates.

Part nine. Twenty pearls.

The first ten pearls cover the disease up to the operation. Biliary atresia is present at birth, because the direct bilirubin is already raised in the first forty-eight hours. Its incidence is about one in five to ten thousand births in East Asia and one in fourteen to nineteen thousand in Europe. The splenic malformation form makes up about ten percent of cases in the West and under three percent in Asia, and it carries worse clearance and more hepatopulmonary syndrome. Newborn direct bilirubin screening found seven of seven cases and moved the mean Kasai from fifty-six to thirty-six days. MMP-7 had an area under the curve of zero point nine and a negative predictive value of ninety-seven percent in the ChiLDReN study, but its cut-off depends on the assay used. Hepatobiliary scintigraphy is sensitive, at almost ninety-nine percent, but its specificity is only seventy percent, and it never replaces the cholangiogram. There is no safe and no futile age: native liver survival at twenty-five years is thirty-eight percent after a first-month Kasai and nineteen percent after three months. Volume is a treatment: sixty-one percent versus fourteen percent at five years. The English programme achieved a median Kasai at fifty-one days, sixty-one percent clearance and fifty-one percent native liver survival at five years. And in the START trial, steroids gave drainage in fifty-nine percent versus forty-nine percent, which was not significant, and doubled serious adverse events in the first month.

The second ten pearls cover life after the operation. Intravenous immunoglobulin, maralixibat and odevixibat have all failed. Cholangitis affects half or more of children, with fever in only four in ten episodes, and is treated for ten to fourteen days when suspected and fourteen to twenty-one days when confirmed. A bilirubin below two at three months gives eighty-six percent two-year transplant-free survival, against twenty percent. The Japanese registry shows native liver survival falling from about fifty percent at ten years to forty-one percent at thirty. About half of survivors have portal hypertension, and most infants with high-risk varices have normal platelets. The hepatopulmonary syndrome is found in about two percent in the Japanese registry and about seven percent where children are actively screened, and portopulmonary hypertension appears mainly in adolescence. Infants need at least one hundred and thirty percent of their energy requirement, with MCT making up thirty to sixty percent of fat but always less than eighty percent, and almost none of the jaundiced infants are sufficient in all vitamins. Living donor grafts did better than split grafts, ninety-four versus eighty-one percent at five years, and the veins must be planned in the splenic malformation. Adults with atresia on the American waiting list had nearly eleven times the waiting list mortality of adolescents. And tumours in the native liver split by age, with hepatocellular carcinoma at a median age of four and cholangiocarcinoma at a median age of about twenty-three and a half.

Part ten. What the block could not answer.

Seven gaps stand out. We do not have universal newborn screening in North America or most of Europe, and the median Kasai outside screening programmes is still forty-eight to sixty-one days; a multistate direct bilirubin programme with audited recall would close that gap. We have no society-endorsed minimum caseload, although every cohort that measured it found an effect of volume; a national audit with outcomes reported by centre would settle it. We have no effective adjuvant drug, and ursodeoxycholic acid, the most widely used drug after the Kasai, has never been tested against placebo; trials stratified by age and phenotype, and a randomised withdrawal trial of ursodeoxycholic acid, are needed. Cholangitis prophylaxis rests on consensus alone, while the meta-analyses of 2023 and 2026 found no fall in cholangitis, so a pragmatic randomised trial of prophylaxis against none is overdue. For varices in infants we have one banding trial of twenty-two children and no beta-blocker trial. Every threshold for the lung syndromes is borrowed from adults, and a registry-based screening schedule is needed. And although fifty-seven tumours in native livers have been published, no guideline tells us how to survey adults; an international adult atresia registry would help.

Part eleven. Ten questions for the mirror.

A short self-test helps to fix the block. If a breastfed three-week-old has a direct bilirubin of one point two and the parents say the stools are yellow, the answer is an urgent work-up this week, and you should look at the stool yourself. A normal gallbladder on ultrasound does not exclude atresia, so the work-up continues with MMP-7, biopsy or cholangiogram. A ninety-five-day-old with no ascites and an INR that corrects with vitamin K should have a Kasai, not a primary transplant. Routine steroids after the Kasai are not recommended, because the START trial showed no benefit and earlier harm. Cholangitis confirmed on criteria with a negative blood culture is still treated, for fourteen to twenty-one days. A bilirubin of three and a half at three months means a two-year transplant-free survival of about twenty percent and a referral for transplant evaluation. A cleared infant with a platelet count of one hundred and sixty still needs endoscopy if portal hypertension is suspected, because normal platelets often hide high-risk varices. A child with polysplenia and a saturation of ninety-seven percent has not had the hepatopulmonary syndrome excluded; a contrast echocardiogram and a blood gas are needed. MCT at eighty-five percent of fat is too much and risks essential fatty acid deficiency. And a new biliary stricture in an adult with a native liver is cholangiocarcinoma until proven otherwise.

Part twelve. What to do differently in clinic.

The block ends with ten changes in practice. Treat every direct bilirubin of one milligram per decilitre or more as urgent, and every pale stool as a same-day referral. Run the GGT, MMP-7, ultrasound and alpha-1 antitrypsin level in parallel, and book the cholangiogram as the first step of the Kasai rather than as a separate test. Send every suspected infant to a centre that does many Kasai operations and transplants. Offer the Kasai at any age unless the liver is decompensated. Give no routine steroids after the Kasai, and use new drugs only within trials. Teach parents the signs of cholangitis, and treat on clinical and laboratory criteria rather than waiting for a culture. Check the bilirubin at three months, and start transplant planning when it is two or above. Feed to at least one hundred and thirty percent of energy needs, measure each vitamin separately, and follow the arm circumference and not only the weight. Screen every survivor every year for varices, the hepatopulmonary syndrome and portopulmonary hypertension, whatever the bilirubin. And hand each young person over to adult care with a written plan that covers pregnancy, cancer surveillance and the triggers for transplant.

The Kasai is the first operation in biliary atresia, not the last decision. The first clock is won or lost in the weeks after birth, mostly by the system around the infant. The second clock never stops, and the job of the hepatologist is to keep listening to it for the rest of the patient's life.